

vasic-digital tmux — Project Constitution

Field	Value
Revision	3
Created	2026-05-07
Last modified	2026-05-21
Status	active
Extends	constitution/Constitution.md — Helix Universal Constitution, pinned 7f738df
Status summary	R3 (2026-05-21): full refactor to the HelixConstitution extends-template form. Universal clauses now inherited from the constitution/ submodule; this file holds only Project Articles §101–§109. No universal clause weakened — inheritance is verified by scripts/tests/test_constitution_inheritance.sh + its paired mutation.

This constitution **extends** the Helix Universal Constitution at constitution/Constitution.md. Every clause there — the anti-bluff covenant (§11.4), absolute data safety (§9), the 60 % host-memory budget, the continuation-document invariant (§12.10), the no-guessing mandate (§11.4.6), commit/push discipline, credentials handling — applies to this repository **unconditionally**.

The Project Articles below **extend** those universal clauses with tmux-specific rules. They never weaken or override a universal clause. **When this file disagrees with the constitution submodule, the constitution wins.**

Inheritance is verified, not assumed. scripts/tests/test_constitution_inheritance.sh asserts the submodule is present at the pinned revision and that the §11.4 *End-user Quality Guarantee* anchor physically exists in constitution/Constitution.md. Its paired mutation — constitution/meta_test_inheritance.sh, wired as the CM-CONSTITUTION-INHERITANCE entry in scripts/tests/meta_test_false_positive_proof.sh — deletes that anchor and proves the gate then FAILs. The gate cannot itself be a bluff gate.

Project Articles

§101 — Anti-bluff covenant binding (origin anchor)

This repository is one of the historical origins of the Helix anti-bluff covenant. The verbatim user mandate that the universal §11.4 codifies is preserved here for forensic continuity — it **MUST** remain in this project's Constitution regardless of inheritance:

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!"

The bar for shipping is "users can use the feature," not "tests pass." Every PASS in this codebase — unit test OR HelixQA Challenge — **MUST** carry positive runtime evidence captured *during execution* that the feature works for the end user. Metadata-only PASS, configuration- only PASS, "absence-of-error" PASS, and grep-only PASS are critical defects regardless of how green the summary line is. **FAIL-bluffs** (a test that exits FAIL for a script-internal reason, not a product defect) are equally forbidden and **MUST** be fixed at the source layer.

The full covenant — recorded-evidence requirement, FAIL-bluff prohibition, no-guessing mandate, test-interrupt-on-discovery — is universal §11.4.1 through §11.4.6. This article binds this project to it and adds the project-specific articles below.

§102 — Operator-path test coverage (project rule, User mandate 2026-05-13)

Forensic anchor: Fixed.md A12 + A13. Tests 11 and 14 reported GREEN while their operator-facing equivalents (tmx new -s X) were broken — test 11 only exercised the explicit-socket path; test 14 hand-spawned systemd-run --user --scope units while the real tmx new workflow placed all sessions in one shared cgroup.

The mandate. Every gate test for a feature **MUST** exercise the SAME entry point an end-user would invoke in production. Tests that bypass the operator's wrapper / helper / install path and reproduce its effects with hand-crafted equivalents **DO NOT** satisfy the universal §11.4.2 captured-evidence requirement. When the operator's path and the test's path diverge:

1. The test header **MUST EXPLICITLY** name the divergence (e.g. "test invokes tmux directly; operator path is tmx new").
2. A **SEPARATE** end-to-end test **MUST** close that divergence with captured evidence on the operator-facing entry point.

Operative test: for every test under scripts/tests/, ask "would an operator hit this code path in their normal workflow?" If no, that test is supplementary; the operator-path test must exist alongside it.

No grep-on-script-content alone. A grep on a script for a literal string is allowed AS A STATIC CHECK *in addition to* a runtime readback — never as the only assertion.

Layer-4 mutations MUST target the operator-path code. Mutations target `scripts/tmx-vm / scripts/tmx.template / scripts/tmux.conf.template` (the code that runs in production), not synthetic-test code.

Non-compliance is a release blocker.

§103 — Four-layer test coverage for tmux defects (User mandate 2026-05-06)

Every defect found, fixed, or introduced in this repo MUST land all four layers before the cycle closes:

- **Layer 1 — pre-build / static-source gate.** A check in `scripts/verify.sh` (or the relevant build script) that catches the defect class at source.
- **Layer 2 — runtime test.** A numbered test in `scripts/tests/NN_*.sh` producing positive runtime artifacts, anti-bluff per §101, operator-path per §102, topology-dispatched per §104.
- **Layer 3 — HelixQA Challenge.** An entry in `scripts/challenges/tmux.yaml` carrying captured-evidence semantics. A test without a paired Challenge is a §101 PASS-bluff.
- **Layer 4 — paired mutation.** A mutation in `scripts/tests/meta_test_false_positive_proof.sh` proving the Layer-2 test catches its own negation (mutation CAUGHT) and that the feature is restored afterwards (RESTORED).

Test-interrupt-on-discovery. The moment any defect is rediscovered, re-produced, or newly identified during a test cycle, the cycle STOPS. Fix at root cause → land all four layers → full rebuild → repeat the full cycle from the beginning.

§104 — Per-host-topology test dispatch for tmux (User mandate 2026-05-05)

Tests that depend on host topology (systemd version, cgroup v1 vs v2, distro-specific controller availability, presence of `systemd-run --user`, Linux vs Darwin) MUST detect topology at test entry and dispatch the topology-appropriate variant — recording the probe output as positive evidence of the SKIP rationale. A test running the wrong variant for the actual topology and PASSing is a §101 PASS-bluff. Silent degradation to a no-isolation path is forbidden.

`scripts/tmx` classifies the host as `tmx-supported / tmx-degraded / tmx-unsupported` via `_probe_topology()` — the canonical dispatch seam.

§105 — Per-session isolation architecture (default since 2026-05-13)

Each `tmx new -s NAME` spawns its OWN tmux server on socket `tmx-NAME` inside its OWN OS-native isolation boundary:

- **Linux** — `cgroup-v2` transient scope `tmx-NAME.scope` via `systemd-run --user --scope`. `MemoryMax` host-adaptive ($\max(\text{MemTotal} \times 60 \% / 4, 2$

GB) unless `TMX_MEM` overrides), `CPUQuota=200%`, `TasksMax=4096`, `Delegate=yes`. OOM in any one scope affects ONLY that scope — `user.slice` survives.

- **macOS (Darwin)** — POSIX `rlimit` wrapper as session `default-command`: `RLIMIT_CPU + RLIMIT_NPROC`, kernel-enforced. `RLIMIT_AS` (memory) is NOT enforced by XNU for unprivileged processes — documented honestly per §101, see `docs/guide/README.md §5.6`.

The session shell is the operator's host shell with full `$PATH`. Plain-vanilla tmux UX with OS-native isolation as a safeguard.

§106 — `TMX_MEM` memory-budget enforcement (project enforcement of universal §12.6)

The universal 60 % host-memory budget is enforced in this project by the per-session containerization in `scripts/tmx`. Invariant:
 $\Sigma(\text{active } \text{TMX_MEM}) \leq 0.6 \times \text{MemTotal}$. Operator default `TMX_MEM=8G`. On Linux the cgroup `MemoryMax` is kernel-enforced; on macOS the cap is informational only (XNU gap, §105). There is NO operator-facing override of the per-host total.

§107 — Upstream tmux submodule pinned + immutable

`tmux/` references `tmux/tmux` at tag 3.6a (commit `cc117b5`). It is a third-party upstream submodule: **never modified**, never advanced off the tag without an explicit, documented decision in `Fixed.md` + a governance-doc update in the same commit. Silent pin-drift is a §101 / universal §11.4.6 violation (forensic precedent: `Fixed.md §3.8`).

§108 — Native dual-OS build discipline

The hardened tmux binary is built natively per OS — Linux ELF and macOS Mach-O — from the same `tmux/` submodule. `scripts/setup.sh`, `scripts/install_deps.sh`, and every project script MUST detect host OS and apply the correct action out of the box (no manual per-OS flags). Compile hardening, jemalloc linkage, and the verification gate apply to both targets.

§109 — Anti-bluff verification gate (load-bearing)

`scripts/verify.sh` is the single decision point for whether the binary is operator-safe. GREEN → `setup.sh` proceeds to `PATH` export; RED → `setup.sh` REFUSES to install. There is no "force install" mode and no flag that bypasses the gate.

Overrides of the Universal Constitution

None. This project introduces **no override** of any universal clause. Any future override MUST appear here under an explicit `### Override §X.Y – <reason>` heading with justification.

Owned-submodule set

Submodules this project owns and is responsible for:

Containers `git@github.com:vasic-digital/Containers.git`
(cgroup orchestration helpers)
constitution `git@github.com:HelixDevelopment/`
`HelixConstitution.git` (governance; pinned 7f738df)

tmux/ is a **third-party upstream** submodule (tmux/tmux tag 3.6a) — excluded from the owned set; immutable per §107.

Project-specific remotes

Repo	Remotes	Push mechanism
Main — vasic-digital/tmux	github + gitlab	bash <code>commit_all.sh "<msg>"</code> — ONLY allowed mechanism
Containers	github + gitlab	the Containers repo's own commit wrapper

Never `git push` directly on the main repo. SSH remotes only — no HTTPS for Git. No CI/CD pipeline files. Force-push is never automatic and requires explicit per-operation authorization per universal §9.

§11.4.87–98 — universal anchors inherited from constitution 6828ff2 (cycle 2026-05-28)

The constitution submodule was ff-merged from 84c948d to 6828ff2 (2026-05-28). Anchors §11.4.87 through §11.4.98 are universal additions that apply to this project unconditionally per §11.4.35. Long-form bodies live in `constitution/Constitution.md`; short-form mirrors follow.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing

Universal anchor inherited from `constitution/Constitution.md` §11.4.87 (commit 6828ff2). Operative for this project: agent work continues autonomously until ALL of (a) `Issues.md` has no In-progress / Ready / In-testing / Reopened items, (b) `CONTINUATION.md` §3 active-work empty, (c) no in-flight subagent, (d) no external dependency in-flight. Every closed item carries §11.4.4(b) four-layer coverage + captured evidence per §11.4.5/§11.4.69. Tests AND HelixQA Challenges bound equally. **Canonical authority:** constitution submodule `Constitution.md` §11.4.87 (6828ff2).

§11.4.88 — Background-push: commit-lock release immediately, push detached

Universal anchor inherited from constitution/Constitution.md §11.4.88 (commit 6828ff2). Operative: `commit_all.sh` MUST release its commit flock the moment `git commit` returns 0; `git push` runs detached via `nohup ... & + disown`; per-remote `.git/.push.<remote>.lock` serialises same-remote concurrent pushes but allows different-remote parallelism. Backgrounded push failures land under `qa-results/push_failures/`. The only escape is `--sync-push` for §11.4.41 force-push paths. **Canonical authority:** constitution submodule Constitution.md §11.4.88 (6828ff2).

§11.4.89 — Background test execution (long tests run detached)

Universal anchor inherited from constitution/Constitution.md §11.4.89 (commit 6828ff2). Operative for this project's >30 s tests (`bash scripts/tests/run_all.sh`, `meta_test_false_positive_proof.sh`, `test_e2e.sh`, etc.): MUST spawn via `nohup ... > <log> 2>&1 & + disown`; main work stream returns to the §11.4.42 priority queue; hard-dependent next steps poll exit-status. Per-script flock serialises same-script invocations. **Canonical authority:** constitution submodule Constitution.md §11.4.89 (6828ff2).

§11.4.90 — Obsolete status + per-item obsolescence audit

Universal anchor inherited from constitution/Constitution.md §11.4.90 (commit 6828ff2). The §11.4.15 Status closed-set gains terminal value `Obsolete` (→ `Fixed.md`) (orthogonal to `Type`). Reasons drawn from closed vocabulary: `superseded-by-design-change` | `superseded-by-later-mandate` | `feature-removed` | `duplicate-of` | `unsupported-topology`. Every `Obsolete` heading carries `**Obsolete-Details:**` (Since + Reason + Superseding-item

- Triple-check evidence) within 8 non-blank lines. §11.4.23 colorizer adds `cell-status-obsolete` class (light-gray + strikethrough). **Canonical authority:** constitution submodule Constitution.md §11.4.90 (6828ff2).

§11.4.91 — Summary-doc clarity mandate

Universal anchor inherited from constitution/Constitution.md §11.4.91 (commit 6828ff2). Every summary one-liner (`Issues_Summary.md`, `Fixed_Summary.md`, `README doc-link`, `Status_Summary` pages) MUST be self-contained — ≥ 6 words OR ≥ 40 chars naming `SUBJECT + PROBLEM/GOAL`. Forbidden anti-patterns: section labels (`Composes with`, etc.); bare metadata fragments (`Critical`, `Bug`, `In progress`); §-letter alone. Generators MUST extract from H1/H2 heading line per §11.4.54 ATM-NNN convention and refuse anti-pattern rows. **Canonical authority:** constitution submodule Constitution.md §11.4.91 (6828ff2).

§11.4.92 — Multi-pass change-evaluation discipline

Universal anchor inherited from constitution/Constitution.md §11.4.92 (commit 6828ff2). Every non-trivial change passes 5 evaluation passes BEFORE commit-ready: (1) main-task captured-evidence verification; (2) regression blast-radius enumeration of every dependent file; (3) cross-feature interaction analysis (shared state/timing/hardware); (4) deep-research per §11.4.8 + CodeGraph queries per §11.4.78/§11.4.79; (5) anti-bluff confirmation. Per-pass written documentation required. Trivial exemption only for zero-source-touch revision-bump / typo commits citing the exemption. **Canonical authority:** constitution submodule Constitution.md §11.4.92 (6828ff2).

§11.4.93 — SQLite-backed single source of truth for workable items

Universal anchor inherited from constitution/Constitution.md §11.4.93 (commit 6828ff2). Text-based Issues/Fixed/Summary/CONTINUATION constellation converts to SQLite-DB-backed single source of truth at docs/workable_items.db (TRACKED per §11.4.95, not gitignored). A Go binary at cmd/workable-items/ provides bidirectional sync (md-to-db / db-to-md / diff / validate / add / close). Round-trip MUST be byte-identical within whitespace-and-section-order tolerance. commit_all.sh refuses to commit while diff is non-empty. **Canonical authority:** constitution submodule Constitution.md §11.4.93 (6828ff2).

§11.4.94 — Zero-idle priority-first parallel-by-default operating mode

Universal anchor inherited from constitution/Constitution.md §11.4.94 (commit 6828ff2). Binds §11.4.20+§11.4.42+§11.4.58+§11.4.70+§11.4.72+§11.4.82+§11.4.87+§11.4.88+§11.4.89 into a single always-on enforcement: idle ONLY when every queued item is genuinely externally blocked OR operator STOP OR §12 host-safety; before any wake/sleep the conductor MUST survey parallel-work feasibility, identify non-contending items, dispatch in parallel per §11.4.20/§11.4.70 subagent-driven default + §11.4.58 PWU disjoint scope + §11.4.89 background long tests. **Canonical authority:** constitution submodule Constitution.md §11.4.94 (6828ff2).

§11.4.95 — Workable-items SQLite DB TRACKED in git, NEVER gitignored

Universal anchor inherited from constitution/Constitution.md §11.4.95 (commit 6828ff2). Amends §11.4.93: docs/workable_items.db IS authoritative source data, NOT a build artefact — it is TRACKED in git, NEVER gitignored. Every workable-items sync md-to-db mutation stages + commits + pushes the DB alongside the MD regen per §11.4.19 atomic-move + §2.1 multi-upstream push. WAL-checkpoint (PRAGMA wal_checkpoint(TRUNCATE)) required before commit-stage. Destructive DB ops require §9.2 hardlinked-backup. **Canonical authority:** constitution submodule Constitution.md §11.4.95 (6828ff2).

§11.4.96 — Safe-parallel-work-with-long-build catalogue + mandate

Universal anchor inherited from constitution/Constitution.md §11.4.96 (commit 6828ff2). Operational catalogue for long-running build workloads classifies parallel work as SAFE (docs/scripts/tests/submodule pushes under §11.4.88 + research + DB ops + subagent dispatch) vs UNSAFE (destructive git ops on source tree, mass deletes under hot subtrees, container destruction, host-safety breaches). Conductor MUST consult catalogue at every pause point and dispatch ≥ 1 safe item per §11.4.20/ §11.4.70 subagent default + §11.4.89 background. **Canonical authority:** constitution submodule Constitution.md §11.4.96 (6828ff2).

§11.4.97 — Maximum-use-of-idle-time + progress-update cadence

Universal anchor inherited from constitution/Constitution.md §11.4.97 (commit 6828ff2). Strengthens §11.4.87+§11.4.94+§11.4.96: every idle minute during which work could autonomously progress and is not genuinely blocked = §11.4.97 violation. Progress-update cadence: emit operator-facing 1-line update at every commit landed / subagent return / constitutional anchor landing / captured-evidence acquisition / milestone closure — no operator prompt required. Continuous physical-proof gathering per §11.4.5+§11.4.6+§11.4.69. **Canonical authority:** constitution submodule Constitution.md §11.4.97 (6828ff2).

§11.4.98 — Full-Automation Anti-Bluff — live tests re-runnable end-to-end without manual intervention

Universal anchor inherited from constitution/Constitution.md §11.4.98 (commit 6828ff2). Closes the manual-intervention gap §11.4.85/§11.4.87/ §11.4.89/ §11.4.94 did not explicitly forbid: every test (unit / integration / e2e / Challenge / stress / chaos / live) MUST be fully self-driving end-to-end and report PASS/FAIL/SKIP-with-reason without any further human action after startup. One-time credential bootstrap OUTSIDE test execution is the only permissible exception. Re-runnability proof = PASS at `-count=3` consecutive automated invocations with self-cleaning state. Manual-dependency tests not rewritten within 30 days graduate to §11.4.90 Obsolete citing §11.4.98 as the reason. **Canonical authority:** constitution submodule Constitution.md §11.4.98 (6828ff2).

This Project Constitution governs the vasic-digital/tmux repository. It is subordinate to and extends constitution/Constitution.md.